EAT: A Benchmark for Annotation Transfer with Raw pLM Embeddings

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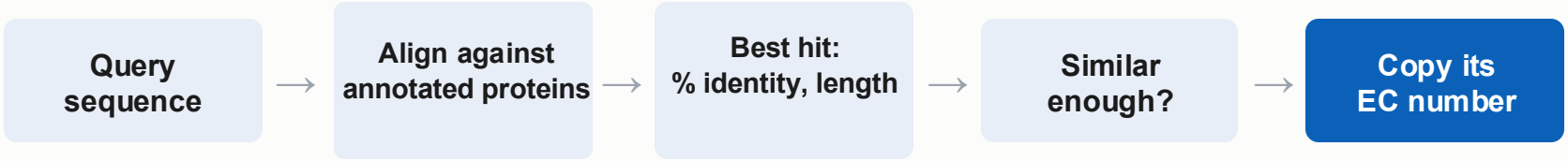
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Today: copy the label from the most similar sequence

~150,000,000 sequences in UniProtKB, only **48,401** enzymes with experimentally verified activity (≈ 0.03 %) [1]. Everything else is annotated by copying from a studied relative.



What "similar" means

% identical residues in an alignment that allows insertions, deletions and conservative substitutions.

Why it works

Similar sequences usually share an ancestor, and with it structure and function.

[1] UniProt Consortium, Nucleic Acids Res (2025); UniProtKB 2026_03: sequences queried 28.10.2026, enzymes = (ec:*) AND (cc_catalytic_activity_exp:*).

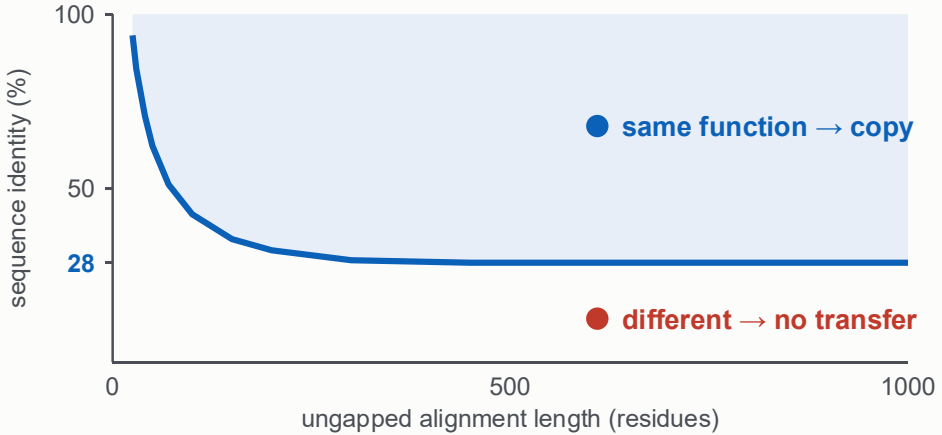
How similar is similar enough? HFSP gives a calibrated answer

One curve over identity and alignment length [11], fitted on Swiss-Prot enzymes [5].

Above it: copy the label. Long alignments need ≥ 28 % identity, short ones more [5].

Held-out test: 96 % precision at EC level 3 [5].

A yes/no with a known error rate.



[5] Mahlich … Bromberg, HFSP, Bioinformatics (2018). Curve redrawn schematically from Eq. 4 / Fig. 1A. [11] B. Rost. Twilight zone of protein sequence alignments. Protein Engineering 12(2):85–94 (1999).

Where copying by sequence fails

< 30 %

Relatives become invisible [11, 2, 3]

Alignments can't separate same from different function.

96 % → 44 %

Precise only at coarse levels [5]

Only 44 % correct at the full EC number.

≥ 1 in 6

Errors get copied too [5]

Swiss-Prot computational annotations flagged wrong by HFSP.

→ Embeddings find relatives that alignments miss (64 % vs 35 % [3]). Copy in embedding space instead?

[2] Littmann … Rost, Sci Rep (2021). [3] Heinzinger … Rost, NAR Genom Bioinform (2022): CATH superfamily level. [5] Mahlich … Bromberg, HFSP, Bioinformatics (2018). [11] B. Rost, Twilight zone of protein sequence alignments, Protein Eng (1999).

EAT works, but it has no HFSP

69.6 → 78.5 %

The pLM matters [12]

CATH superfamily accuracy, ProtT5 → ProstT5.

64 % vs 73 %

No shared protocol [3, 4]

Same ProtT5, same CATH sets, two papers.

RI = 0.5/(0.5+d)

No calibrated cutoff [2]

Cutoffs set for SeqVec; other pLMs' distances differ in scale.

→ We build it: a raw-EAT benchmark on CATH and EC. Nothing trained, so pLMs compare fairly and any new method can be checked against it.

[2] Littmann … Rost, Sci Rep (2021). [3] Heinzinger … Rost, NAR Genom Bioinform (2022). [4] Elnaggar … Rost, Ankh, arXiv (2023). [12] Miller … Orengo, ContrasTED, bioRxiv (2026).

Approach: Embedding-based annotation transfer (EAT)

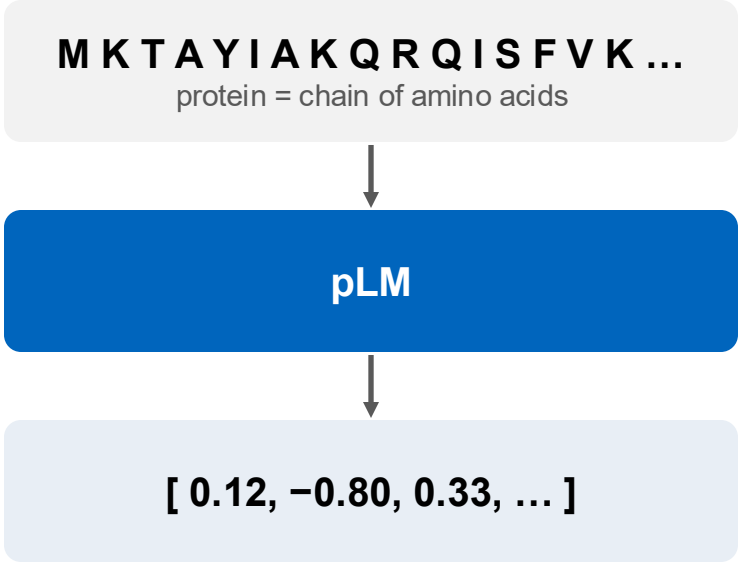


• Without any training, an early version came close to the top 10 of a worldwide prediction contest (CAFA3) [2]

• Embedding-based methods can identify distant structural relationships more accurately than traditional sequence search. [3]

[2] M. Littmann … B. Rost. Embeddings transfer GO annotations beyond homology. Sci Rep (2021).
[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

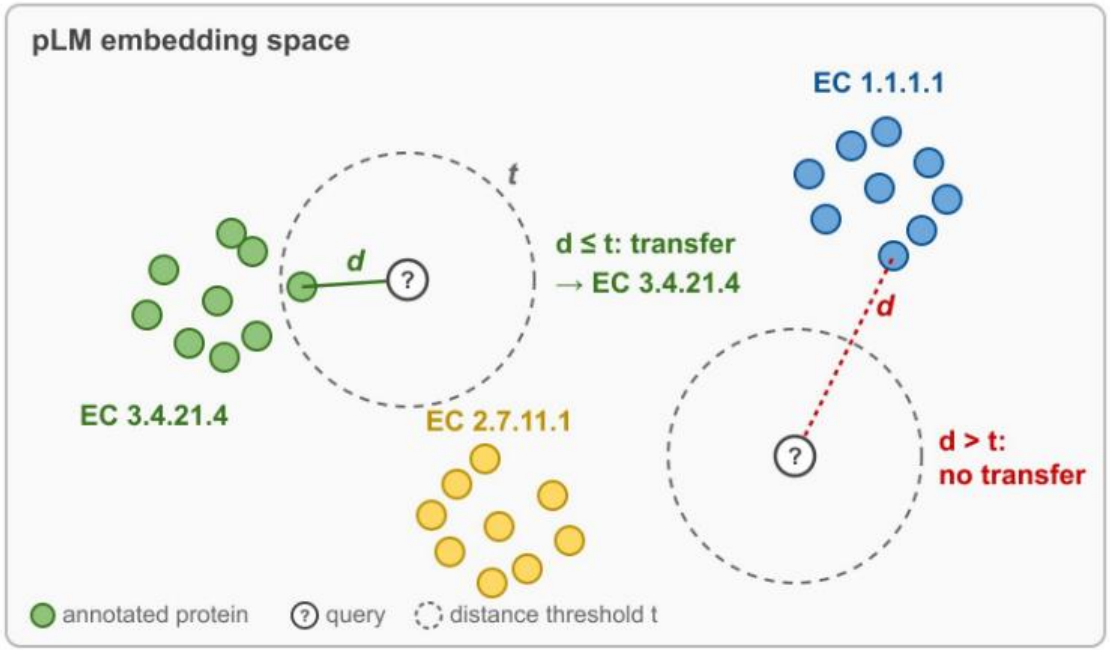
pLMs – Protein Language Models



- pLMs encode proteins into a vector space
- structurally / functionally similar proteins typically end up close together
- different pLMs have different objectives and embedding spaces.

[8] A. Elnaggar … B. Rost. ProtTrans. IEEE TPAMI (2022). [9] Z. Lin … A. Rives. ESM-2. Science (2023). [4] Elnaggar … Rost. Ankh (2023).

Embedding-based Annotation Transfer (EAT)



- **Nearest neighbour:** the closest protein with a known label
- **Distance:** straight line (Euclidean) or angle between vectors (cosine)
- **Threshold:** copy label if below, model-specific

[2] Littmann … Rost (2021).

Labels – CATH and EC

CATH: the 3D shape of a domain

domain = building block of a protein

3 Class

3.40 Architecture

3.40.50 Topology

3.40.50.300 Homologous Superfamily

EC: the reaction an enzyme performs

enzyme = protein that speeds up a chemical reaction

3 Class

3.4 Subclass

3.4.21 Sub-subclass

3.4.21.4 Enzyme

→ measure accuracy and thresholds per level: up to which level does copying still work?

[6] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021). [7] IUBMB. Enzyme Nomenclature (ExplorEnz).
Data: [3], [5], UniProt [1].

Research Questions

Q1: Which pLM and which distance measure (e.g., Euclidean or cosine) transfer structure and function annotations best, and where does EAT beat plain sequence similarity?

Q2: Up to which distance is a transfer safe (e.g., ≥95% precision), for structure and for function?
Can distance be turned into a reliability score that means the same for every pLM (e.g., 0.9 = 90% of such transfers are correct)?

Approach

1 Reproduce

Rerun a published result: ProtT5 on CATH and verify

2 Compare

Compare models

Compare EAT vs sequence similarity

3 Homogenize

Both CATH and EC

Distances and thresholds per model

Freeze splits

Lookup set

Tuning set

Test set

[3] Heinzinger … Rost (2022): published ProtT5 result on CATH test219.

References

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025).

[2] M. Littmann … B. Rost. Embeddings from deep learning transfer GO annotations beyond homology. Sci Rep (2021).

[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

[4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model unlocks general-purpose modelling. arXiv (2023).

[5] Y. Mahlich … Y. Bromberg. HFSP: High speed homology-driven function annotation of proteins. Bioinformatics (2018).

[6] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021).

[7] IUBMB. Enzyme Nomenclature. ExplorEnz database, www.enzyme-database.org.

[8] A. Elnaggar … B. Rost. ProtTrans: Toward understanding the language of life. IEEE TPAMI (2022).

[9] Z. Lin … A. Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science (2023).

[10] Ribeiro et al. (2020), The FEBS Journal, Table 1.

[11] B. Rost. Twilight zone of protein sequence alignments. Protein Engineering 12(2):85–94 (1999).

[12] Miller … Orengo, ContrasTED, bioRxiv (2026).

Project description: Chair of Bioinformatics (I12), TUM. EAT: A benchmark for annotation transfer with raw pLM embeddings (2026).